HANSA THARUKA

Senior Full Stack Engineer | Virtusa Certified AI Native Engineer
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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 6+ years at Virtusa building and supporting enterprise applications for global clients in the Insurance and Human Resources (Compensation) domains. Virtusa Certified AI Native Engineer, skilled in using AI tools across the development lifecycle to speed up coding, testing, debugging and documentation. Delivers end-to-end features with Angular, Next.js, NestJS, Node.js, Python, Java Spring Boot and MuleSoft, backed by MongoDB, Redis and Oracle. Experienced in AI agent services (Mastra), microservices, secure API gateways, platform migrations, CI/CD, AWS, Docker and Kubernetes.

TECHNICAL SKILLS

Frontend: Angular 20, Next.js, TypeScript, JavaScript, HTML5, CSS3

Backend: NestJS, Node.js, Python, Java, Spring Boot, Spring Cloud, REST, SOAP, Microservices

AI and Data: Mastra (AI agents), AI-assisted development, Databricks

Integration: MuleSoft (Anypoint, API Manager, Runtime Fabric), DataWeave, Apache Kafka, AWS MSK

Databases: MongoDB, Redis, Oracle DB, MySQL

Cloud and DevOps: AWS (EKS, MSK, EC2), Docker, Kubernetes, Jenkins, GitLab CI, Bamboo, Nx, Maven, Git

Testing: Jest, Cypress, JUnit, Selenium, ReadyAPI, Postman

Monitoring: Datadog, Komodor, Sumo Logic, New Relic, Grafana, Kibana, ELK, Splunk

Security: OAuth 2.0, JWT, TLS/SSL, AES-256, HashiCorp Vault, CyberArk Conjur, Wiz, OWASP

PROFESSIONAL EXPERIENCE

Senior Software Engineer (Full Stack) — Virtusa (Pvt) Ltd, Sri Lanka

2020 – Present

Virtusa Compensation Assistant – AI (Human Resources)

Tech: Angular 20, Next.js, NestJS, Node.js, TypeScript, Python, Mastra, MongoDB, Redis, Databricks, Nx, Jest, Cypress, Datadog, Komodor

- Built features for an AI-powered compensation market-pricing platform (Nx monorepo, Angular 20 frontend, NestJS/MongoDB API, Mastra agent service) automating benchmarking, peer-group resolution and regression-based pay recommendations.
- Developed and fixed export features for benchmark charts and regression results, updating export endpoints, file naming and pay-component values, and fixing the missing negative sign for below-market figures.
- Fixed defects in the default benchmarking stage, including title casing, data card expansion and pay-component ordering, improving the accuracy of displayed results.
- Addressed code-review feedback, wrote and repaired Jest unit tests, updated Swagger documentation, fixed build and formatting issues, and upgraded dd-trace to clear a Wiz security finding.
- Managed deployments across environments and monitored performance with Datadog and Komodor, identifying bottlenecks and shipping fixes to keep the platform stable.

Cross Tech (Insurance)

Tech: Java, Spring Boot, MuleSoft, Runtime Fabric, CyberArk Conjur, Sumo Logic, AWS

- Resolved production issues reported by business support in MuleSoft and Spring Boot applications end to end: root-cause analysis, risk analysis, fixes in lower environments, QA handover, production release and post-release validation.
- Designed, built and tested urgent new components and projects from scratch through to production release and support.
- Ran impact and risk analysis for fixes and new work, keeping stakeholders and cross-functional teams informed.

- Coordinated QA testing, released to production and monitored post-release performance.

Vault to Conjur Migration (Insurance)

Tech: Java, Spring Boot, MuleSoft, MuleSoft Runtime Fabric, CyberArk Conjur, HashiCorp Vault, WildFly, Sumo Logic, AWS, Grafana, Kibana

- Planned and executed the migration of secure properties from HashiCorp Vault to CyberArk Conjur for Spring Boot and MuleSoft services, extending the configuration service to integrate with Conjur.
- Built reusable Java libraries for Spring Boot applications, including a WildFly-specific variant, to load configuration at startup and reduce startup time.
- Wrote the migration plan with impact analysis and risk assessment, coordinating teams to minimise downtime.
- Built Grafana and Kibana dashboards to monitor migrated services, and handed over services with documentation and knowledge-transfer sessions.

External Gateway (Insurance)

Tech: MuleSoft, MuleSoft Runtime Fabric, Java, Shell Script, Sumo Logic, Consul, HashiCorp Vault, AWS EKS

- Developed an API gateway with MuleSoft and Java for secure, efficient communication and data integration, securing APIs with OAuth 2.0, JWT and API-key authentication.
- Configured rate limiting, throttling and traffic shaping with MuleSoft API Manager.
- Applied TLS/SSL encryption in transit and AES-256 encryption at rest.
- Set up ELK Stack, Splunk and Anypoint Monitoring with real-time alerts, and ensured OWASP and GDPR compliance through regular security audits and vulnerability assessments.

Foundation (Insurance)

Tech: MuleSoft, Java Spring Boot, Spring Cloud, Bash Script, AWS (EKS, MSK, EC2), Apache Kafka, Kerberos, Sumo Logic, New Relic, Consul, HashiCorp Vault, Jenkins, GitLab CI, Bamboo

- Managed MuleSoft gateway proxies and delivered API integration change requests with thorough deployment testing.
- Maintained Jenkins, GitLab CI and Bamboo jobs and scripts, reducing deployment time and errors.
- Migrated the communication API from Apache Kafka to Amazon MSK, improving throughput and reliability.
- Supported external partners (Fineos), internal teams (Global Business Services) and the New York Life team with debugging and detailed log analysis to speed up incident resolution.
- Implemented enhancements in Spring Boot microservices using Spring Cloud, and managed application configuration with Consul and HashiCorp Vault.

WAS Migration (Insurance)

Tech: Java, Oracle DB, IBM WebSphere, WildFly, Consul, HashiCorp Vault, Sumo Logic, New Relic, JUnit, PractiTest

- Analysed legacy code and migrated services from IBM WebSphere to WildFly, upgrading Java from version 6 to 8.
- Built proofs of concept integrating Consul and HashiCorp Vault to load configuration at application startup.
- Fixed post-migration defects and ran functional, JUnit and load tests to remove performance bottlenecks.
- Queried Oracle databases to test payloads and fix data issues, debugged with Sumo Logic, tracked work in PractiTest and reported fixes to clients.

EDUCATION

BSc (Hons) in Information Technology — University of Moratuwa, Sri Lanka (GPA 3.01)

International Diploma in Information Technology — Esoft Metro Campus (Pearson UK / Edexcel)

CERTIFICATIONS

Virtusa Certified AI Native Engineer — Virtusa